

Course Code	21CSE656T	Course Name	SOCIAL MEDIA ANALYTICS	Course Category	E	PROFESSIONAL ELECTIVE	L	T	P	C
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Pre-requisite Courses	Nil	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	School of Computing	Data Book / Codes / Standards	Nil		

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	understand the foundations of Social Media Analytics.
CLR-2:	visualize and understand the data mining aspects of social networks.
CLR-3:	solve mining problems with different algorithms.
CLR-4:	understand network measures for social data.
CLR-5:	understand the behavioral part of web applications for Analysis.

Course Outcomes (CO):	At the end of this course, learners will be able to:	Programme Outcomes (PO)		
		1	2	3
CO-1:	understand the basics of Social Media Analytics.	3	-	-
CO-2:	explain the significance of Data mining in Social media.	-	2	-
CO-3:	demonstrate the algorithms used for text mining.	-	-	1
CO-4:	apply network measures for social media data.	-	3	-
CO-5:	explain Behavior Analytics techniques used for social media data.	-	2	-

Module-1 - Introduction to Social Media Data	9 Hour
Characteristics of social media data: Volume, velocity, variety (the 3 Vs of big data), Structured, semi-structured, and unstructured data, Challenges and opportunities in social media data analysis Social media data sources and APIs: Publicly available datasets (e.g., Twitter API, Reddit API), Scraping and data collection tools (e.g., BeautifulSoup, Selenium), Ethical considerations in data collection (privacy, consent, terms of service) Preprocessing and cleaning social media data: Handling noise, duplicates, and missing values, Text normalization (tokenization, stemming, lemmatization), Feature engineering for social media data (e.g., sentiment scores, network metrics). T1 Text Pre Processing, T2 Sentiment Analysis	
Module-2 - Text Mining and Natural Language Processing (NLP) for social media	9 Hour
Text representation: Bag-of-words, TF-IDF, word embeddings (e.g., Word2Vec, GloVe), Topic modeling (e.g., Latent Dirichlet Allocation), Sentiment analysis and opinion mining: Lexicon-based approaches (e.g., VADER), Machine learning-based approaches (e.g., Naive Bayes, SVM, deep learning), Aspect-based sentiment analysis Text classification and categorization: Supervised and unsupervised approaches, Applications (e.g., spam detection, topic classification). T3 Spam Detection, T4 Topic Classification	
Module-3 - Introduction to Social Networks	9 Hour
Definitions and key concepts: Nodes, edges, attributes, directed/undirected networks, weighted/unweighted networks, Types of networks: Social, information, biological, technological, Historical context and applications of SNA, Network Representation and Visualization, Sociograms and graph layouts (force-directed, circular, hierarchical), Adjacency matrices and edge lists, Network visualization tools, Centrality Measures: Degree centrality: Number of connections, Betweenness, Closeness centrality, Eigenvector centrality. T5 Graph Centrality measures, T6 Network Visualization	
Module-4 - Social Network Structure, Visualization and Analysis	9 Hour

Network Structure and Properties. Density, Reciprocity, diameter, clustering coefficient, Transitivity, Cliques and communities, Community Detection: Modularity-based methods. Spectral clustering. Hierarchical clustering. Network Dynamics, Link prediction, Information diffusion, Multi-Layer Networks, Network visualization tools (e.g., Gephi/ NetworkX), Applications of SNA: Identifying influencers and opinion leaders, Viral marketing and information diffusion, Community detection and segmentation, T7 Influencer Identification, T8 Community Detection

Module-5 - Behavior Analytics for Social Media

9 Hour

User behavior modeling: Clickstream analysis (tracking user interactions with content), Purchase behavior analysis (analyzing buying patterns), Churn prediction (identifying users likely to disengage) Predictive analytics: Recommender systems for social media (suggesting content, products, or connections), Forecasting trends and viral content, Predicting user responses to campaigns Measuring user engagement: Time spent on platform, content interactions (likes, comments, shares), Sentiment analysis of user-generated content, Analyzing user feedback and reviews, T9 – User engagement analysis, T10 – User review analysis

Learning Resources	1. Social Media Analytics in Predicting Consumer Behavior, Selay Ilgaz Sumer, Nurettin Parilti, CRC Press, First Edition, 2023.	4. Analyzing Social Networks Using R, Stephen P. Borgatti, Martin G. Everett, Jeffrey C. Johnson, SAGE Publications, First Edition, 2022.
	2. Social Media Analytics Strategy, Using Data to Optimize Business Performance, April Ursula Fox, First Edition, 2022.	5. Spatializing Social Media, Social Networks Online and Offline, Marco Bastos, Taylor & Francis, First Edition, 2021.
	3. Social Media Analytics and Practical Applications, Liangfei Qiu, Liangfei Qiu, Subodha Kumar, CRC Press, First Edition, 2022.	6. Applied Social Network Analysis with R, Emerging Research and Opportunities, Mehmet Gençer, IGI Global, First Edition, 2020

Learning Assessment							
	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (50%)		Life Long Learning CLA-2 (10%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	10%	-	20%	-
Level 2	Understand	20%	-	10%	-	20%	-
Level 3	Apply	30%	-	40%	-	30%	-
Level 4	Analyze	30%	-	40%	-	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
1.Mr. N. Venkatesh, Tech Lead, Honeywell, Bengaluru, Karnataka, India	1.Dr.V.Sivakumar, Associate Professor, Department of Information and Communication Technology, Manipal Institute of Technology,	1.Dr.T.Veeramakali, SRMIST
2.Ms. Sudha B G, Great Learning		2.Dr.G.Vadivu, SRMIST